



POLITECNICO
MILANO 1863

SYSTEMS AND METHODS FOR BIG AND UNSTRUCTURED DATA

Prof. Marco Brambilla

January 12, 2023

THEORY (6 PTs.)

Choose whether the following statements are true or false by marking the CORRECT answers with an X.
Each correct answer awards 0.5 PTs. Each incorrect answer penalizes -0,125 PTs. Missing answers count 0 PTs.

LAST NAME _____ FIRST NAME _____

	STATEMENT	TRUE	FALSE	Mark
	1 All APIs require the user to get an authentication key to be able to access their services.	T	F	
it's isolation	2 The Consistency (ACID) property ensures that a transaction is not affected by the behaviour of other, concurrent transactions.	T	F	
	3 A graph is sparse when the cardinality of the nodes is similar to the cardinality of the arcs.	T	F	
	4 MongoDB guarantees Partition tolerance and Consistency.	T	F	
should be avoided	5 In Elasticsearch, term queries can be easily applied to text fields	T	F	
	6 HDFS is a distributed, resilient file system built with the objective of supporting scalable computation	T	F	
used also as cache	7 Redis is not meant to be used when caching is needed.	T	F	
	8 In Spark, RDDs are immutable and lazily evaluated.	T	F	
only prescriptive	9 Data-driven decisions can be guided by Descriptive, Predictive, Reactive, or Prescriptive analysis on the data	T	F	
	10 In a documental database, a subdocument is a complex object associated to an attribute in the main document	T	F	
not a DB software, not for queries	11 Map-reduce is a database software that implements scalable queries upon big data	T	F	
	12 A dataframe is a large-scale distributed collection of rows organized into named columns	T	F	



Last Name _____ First Name _____ Codice Persona _____

NASA (National Aeronautics and Space Administration) wants to model a database of their space launches. In particular, they want to manage two types of space systems: spaceships and space probes. A spaceship is described by its name, version, price, number of seats, and fuel capacity. A space probe is described by its name, version, price, and fuel capacity. Each spaceship has an associated crew, while probes are unmanned. Every crew has a unique name, and includes at least three astronauts, up to the number of seats in the spaceship. Each astronaut is described by a personal ID, name, surname, birth date, crew role (engineer, pilot, second pilot, etc.), and hiring date. Each astronaut can be part of several crews with different roles. Each spaceship or space probe can only take part in one launch, and it's assigned to only one crew. Each launch is described by date, time, departure country, departure location, and destination. Destinations are described by their names, coordinates, description, and type. The type depends on the systems: Spaceships head towards: space station, moon landing, planet landing, or orbit only. In contrast, space probes are launched towards space bodies (e.g., planets, satellites, sun, comets, etc.).

Exercise 1 (4 PTs.)

1.1) Model the ER diagram of the proposed situation. Be sure to properly set the primary key for each entity. If needed, write the necessary assumptions to model the diagram properly.

Write the relationship reading direction (3 PTs.).



1.2) Which DB technology would be the best to store the different groups of entities in your ER diagram? For each DB technology, provide the list of entities that would be stored and the motivations (e.g., "Documental DB" - "Entity 1, Entity 2, etc." - "Motivation"). (1 PT)

DB TYPE	MAIN ENTITIES	MOTIVATION

Exercise 2 - Neo4j (4 PTs.)

Consider the following entities: destinations, launches, spaceships, and space probes.

2.1) Provide an example of a simple graph describing the aforementioned nodes, the respective connections, and their properties. Be coherent with your ER Diagram. (1PT)

2.2) Write a Cypher query that computes the total price for each destination of space probes whose individual price is higher than 30 MLN \$, leaving from the country "USA". (1.5 PTs.)

2.3) Write a query to extract the astronauts that traveled at least two times with the same colleague, on trips that departed from the same location and had different type of destination. Make sure not to extract many times the same pairs of astronauts. (1.5 PTs.)

Exercise 3 – MongoDB (4 PTs.)

Consider the following entities: spaceships, crew, and astronauts.

Simplifying assumption: each astronaut can now only be part of one crew.

3.1) Describe the structure of the document you would design to model their relationships and attributes. Be coherent with your ER Diagram. (1 PT.)

3.2) Extract the name of all the astronauts whose spaceship's name is "Dragonstar". (1 PT.)

3.3) Extract the list of all spaceships with a crew of at least five people, one of which must be an engineer hired in 2015 (suppose that the year() function would let you collect the year of a given date). (2 PTs.)

Exercise 4 – Elasticsearch (2 PTs.)

Consider only the entity: destination.

4.1) Describe the **full mapping** in an Elasticsearch model to define the document of the entity involved. Be coherent with your ER diagram, and be sure to assign the field types “text” and “keyword” properly. (1 PT.)

4.2) Write a query to extract all the destinations whose descriptions include the words "travel" and "solar system", assigning a higher score to the ones whose type is "planet" and filtering out the ones with the name "Mars". (1 PT.)

Exercise 5 – Cassandra (2 PTs.)

Consider the following entity: space probe.

5.1) Write the statements to create a keyspace, connect to it, and create a table in Cassandra for the aforementioned entity. Be coherent with your ER Diagram. (1 PT.)

5.2) Write a query to extract the space probes whose version equals "v1.1.0". (1 PT.)